First Working Model: PPE Detection

P14 — Weekly progress report, 8–13 October 2026

Data loading and preprocessing

We use the public Ultralytics Construction-PPE dataset with all 11 original detection classes. Our download script verifies the archive checksum. The preparation script checks image–label pairs and YOLO box coordinates, removes one duplicate box from the prepared copy, and writes a dataset YAML for the training loader. It retains all 1,416 images and 11,520 boxes. To limit leakage, we reviewed similar images across the original splits and grouped related views and transformed copies before assigning them to a single split. The resulting train/validation/test split contains 1,265/73/78 images. The original files remain unchanged; the split manifest records every assignment.

At loading time, images are converted to RGB and resized with aspect-ratio-preserving letterbox padding to 640×640. Training images receive horizontal flips, small scale and translation changes, and mild HSV colour changes. Validation images only receive the deterministic resize and padding. Mosaic, mixup, and copy-paste are disabled.

First working baseline

We fine-tuned COCO-pretrained YOLO11n on all 11 classes for 50 epochs, using batch size 16, SGD (initial learning rate 0.01, momentum 0.937, weight decay 0.0005), three warm-up epochs, and cosine learning-rate decay. Three runs used seeds 42, 43, and 44 on an Apple M4 GPU through PyTorch MPS. Each run took about 50–52 minutes. We evaluated the saved validation-selected checkpoint with class-aware NMS at IoU 0.70 and a confidence cutoff of 0.001 for AP calculation. The test split has not been used for model selection.

Validation run	mAP@0.50:0.95	mAP@0.50	Precision	Recall
Seed 42	0.1895	0.4588	0.5967	0.4632
Seed 43	0.2023	0.4955	0.6290	0.4514
Seed 44	0.1934	0.4737	0.5165	0.4836
Mean ± SD	0.1951 ± 0.0066	0.4760 ± 0.0185	0.5807 ± 0.0579	0.4661 ± 0.0163

Observations and next steps

The baseline detects common categories more reliably than rare ones. For seed 42, AP@0.50 is 0.8394 for `Person` and 0.7211 for `helmet`, but 0.2188 for `goggles` and 0.0599 for `no_boots`. The validation set has only eight `goggles` boxes and four `no_boots` boxes, so those per-class results are unstable. Some source labels are also ambiguous: ordinary glasses and footwear appear in PPE-related classes, and the images are not limited to construction sites. We will compare the planned YOLO11s model on the same split and examine thresholds and failure cases before drawing conclusions from the held-out test set.

Project repository: `https://github.com/Ta6leTkaa/intro-to-CV-Project`. The current code, split manifest, training settings, checkpoints, and per-seed results are available there. The dataset itself is downloaded locally by the provided script.